CLASS-8

LESSON-6 CUBES AND CUBE ROOTS

(This PDF Based on NCERT book)

CUBES(घन)-

Cube means multiplying that number by itself **three times**.

Perfect Cubes (1 to 10)

Here is a list of the first ten perfect cubes:

Number (n)	**Calculation (n×n×n)**	**Cube (n3)**
1	$1 \times 1 \times 1$	**1**
2	$2 \times 2 \times 2$	**8**
3	$3 \times 3 \times 3$	**27**
4	$4 \times 4 \times 4$	**64**
5	$5 \times 5 \times 5$	**125**
6	$6 \times 6 \times 6$	**216**
7	$7 \times 7 \times 7$	**343**
8	$8 \times 8 \times 8$	**512**
9	$9 \times 9 \times 9$	**729**
10	$10 \times 10 \times 10$	**1000**

SOME INTERESTING PATTERS(कुछ दिलचस्प पैटर्न)-

1. **Adding consecutive odd number(लगातार विषम संख्याओ का योग)**

This is one of the most beautiful patterns in arithmetic. Every cube is the sum of a specific set of consecutive odd numbers.

- $1^3 = 1$
- $2^3 = 3 + 5 = 8$
- $3^3 = 7 + 9 + 11 = 27$
- $4^3 = 13 + 15 + 17 + 19 = 64$
- $5^3 = 21 + 23 + 25 + 27 + 29 = 125$

The Rule: To find n^3, you add together n consecutive odd numbers, starting right after where the previous cube left off.

2. **Cubes and their prime factor(घन और उनके अभाज्य गुणनखंड)**

For any number to be a perfect cube, every prime factor in its prime factorization must appear in a multiple of 3 **(3, 6, 9, etc.). Examples of the Pattern**

1. The Cube of 6 ($6^3 = 216$)

First, find the prime factors of 6:

- $6 = 2 \times 3$

Now, look at the prime factors of 216:

- $216 = 2 \times 2 \times 2 \times 3 \times 3 \times 3$
- **Pattern:** We have three 2s (2^3) and three 3s (3^3).

2. The Cube of 10 ($10^3 = 1000$)

First, find the prime factors of 10:

- $10 = 2 \times 5$

Now, look at the prime factors of 1000:

- $1000 = 2 \times 2 \times 2 \times 5 \times 5 \times 5$
- **Pattern:** We have three 2s (2^3) and three 5s (5^3).

How to use this to find a Cube Root

If you have a large number and want to find its cube root using prime factors, you just:

1. Find the prime factorization.
2. Group the factors into sets of three.
3. Pick **one** number from each set and multiply them.

Example: Find the cube root of 1,728

- Prime factors: 2 × 2 × 2 × 2 × 2 × 2 × 3 × 3 × 3
- Group into threes: (2 × 2 × 2) × (2 × 2 × 2) × (3 × 3 × 3)
- Take one from each: 2 × 2 × 3 = 12
- So, 12^3 = 1,728.

Quick Comparison Table

Perfect Cube	Prime Factorization	Power Form
8	2 × 2 × 2	2^3
27	3 × 3 × 3	3^3
64	2 × 2 × 2 × 2 × 2 × 2	2^6
125	5 × 5 × 5	5^3
343	7 × 7 × 7	7^3

SMALLEST MULTIPLE THAT IS A PERFECT CUBE(सबसे छोटा गुणज जो एक पूर्ण गुणज हो)-

To find the **smallest multiple** that makes a number a perfect cube, you just need to finish those "teams of three" we talked about.

Think of it like a game of **"Fill the Gap."** If a prime factor doesn't have 3 partners, you just give it what it’s missing.

How to do it in 3 steps:

1. **Break it down:** Find the prime factors (the building blocks).
2. **Count them:** See how many of each number you have.
3. **Fill the gaps:** Add just enough numbers so that every group has exactly **3** (or 6, 9, etc.).

Example 1: Making 12 a Perfect Cube

1. Break it down:

12 = 2 × 2 × 3

2. Count them:

- We have **two** 2s. (Needs one more to make three).
- We have **one** 3. (Needs two more to make three).

3. Fill the gaps:

We need to multiply by one **2** and two **3s** (3×3).

$2 \times 3 \times 3 = 18$

The Result: The smallest multiple is **18**.

$12 \times 18 = 216$, which is 6^3

Example 2: Making 50 a Perfect Cube

1. Break it down:

$50 = 2 \times 5 \times 5$

2. Count them:

- We have **one** 2. (Needs two more: 2×2).
- We have **two** 5s. (Needs one more: 5).

3. Fill the gaps:

Multiply by $2 \times 2 \times 5 = 20$.

The Result: The smallest multiple is **20**.

$50 \times 20 = 1000$, which is 10^3

Example 2: Making 1,080 a Perfect Cube

1. Break it down (Prime Factors):

$1,080 = 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 5$

2. Check the teams:

- **The 2s:** Three 2s. **Complete!**
- **The 3s:** Three 3s. **Complete!**
- **The 5s:** Only one 5. **Incomplete** (it needs two more 5s to make a team of three).

3. Fill the gap:

We need to multiply by 5×5, which is **25**.

The Result: The smallest multiple is **25**.

Calculation: $1,080 \times 25 = 27,000$ (which is 30^3)

<u>FINDING CUBE ROOTS(घनमूल ज्ञात करना)-</u>

1. The Prime Factorization Method (अभाज्य गुणनखंड विधि)

This is the most reliable way for "perfect cubes." You break the number into its smallest building blocks and group them into **teams of three**.

Example: Find the cube root of 216

1. **Factorize:** $216 = 2 \times 2 \times 2 \times 3 \times 3 \times 3$
2. **Group:** $(2 \times 2 \times 2) \times (3 \times 3 \times 3)$
3. **Pick one from each team:** $2 \times 3 = 6$

   The cube root of 216 is 6.